

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12B.2

EXPERIMENTAL AND THEORETICAL LATTICE ENERGIES

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12B.2: Experimental and Theoretical Lattice Energies
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12B.2

Q1. The theoretical lattice energy of silver iodide (AgI) is -736 kJ mol^{-1} . The experimental (Born-Haber) lattice energy is -889 kJ mol^{-1} . Explain the difference between these two values. [WCH14/01/Oct20/Q2(c)]

[3 Marks] **WCH14/01**

Q2. Which factors increase the polarising power of a cation? [WCH14/01/Jan23/Q4(a)]

[2 Marks] **WCH14/01**

SECTION B: WORKED MARK SCHEME — 12B.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: The theoretical lattice energy of silver iodide (AgI) is -736 kJ mol^{-1}. The experimental (Born-Haber) lattice energy is -889 kJ mol^{-1}. Explain the difference between these two values. [WCH14/01/Oct20/Q2(c)] - The experimental value is more exothermic / negative than the theoretical value (1) - This indicates covalent character in the ionic lattice / polarisation of the iodide ion by the silver ion (1) - The theoretical model assumes a purely ionic model / perfect spheres (1)	[3]
Q2	Q2: Which factors increase the polarising power of a cation? [WCH14/01/Jan23/Q4(a)] - A small ionic radius (1) - A high positive charge (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12B.2**FAQ #1: How do examiners test the concept of polarisation?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

How do examiners test the concept of polarisation?

MODEL ANSWER / EXAMINER EXPECTATION:

Examiners will give you experimental and theoretical lattice energies and ask why they differ. You must state that a large difference implies significant covalent character due to polarisation of the anion by the cation.

FAQ #2: What is the difference between polarising power and polarisability?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the difference between polarising power and polarisability?

MODEL ANSWER / EXAMINER EXPECTATION:

Cations have polarising power (they distort anions). Anions are polarisable (they get distorted). Never mix these up; saying a cation is polarisable will lose you the mark.